

NullPointerException

One.java

CODE

```
package exception;

public class One {

    public static void main(String[] args) {

        try {

            //null is not a string
            String name = null;
            System.out.println(name.length());

        }
        catch (NullPointerException e) {

            System.out.println("Object is null");

        }
    }
}
```

OUTPUT



One.java

> java One

Object is null

ArrayIndexOutOfBoundsException

Two.java

CODE

```
package exception;

public class Two {

    public static void main(String[] args) {

        try {

            //index start from 0
            //max index of array a is 2
            int[] a = {10, 20, 30};
            System.out.println(a[5]);

        }
        catch (ArrayIndexOutOfBoundsException e) {

            System.out.println("Invalid array index");

        }
    }
}
```

OUTPUT

A terminal window with a dark background and a title bar containing three colored circles (red, yellow, green) and the text "Two.java". The terminal shows a green prompt character followed by the command "java Two". Below the command, the output "Invalid array index" is displayed in white text.

```
Two.java
> java Two
Invalid array index
```

NumberFormatException

Three.java

CODE

```
package exception;

public class Three {

    public static void main(String[] args) {

        try {

            //abc is not a number
            String s = "abc";
            int num = Integer.parseInt(s);

        }
        catch (NumberFormatException e) {

            System.out.println("invalid number");

        }
    }
}
```

OUTPUT



Three.java

> java Three

invalid number

StringIndexOutOfBoundsException

Four.java

CODE

```
package exception;

public class Four {

    public static void main(String[] args) {

        try {

            //string index is 4 for java
            String name = "Java";
            System.out.println(name.charAt(10));

        }
        catch(StringIndexOutOfBoundsException e){

            System.out.println(e.getMessage());

        }
    }
}
```

OUTPUT

Four.java

> java Four

String index out of range: 10

ClassCastException

Five.java

CODE

```
package exception;

public class Five {

    public static void main(String[] args) {

        Object obj = "aa";

        try {

            // aa is not a integer
            Integer num = (Integer) obj;
            System.out.println(num);

        }
        catch(ClassCastException e) {

            System.out.println("invalid type casting");

        }
    }
}
```

OUTPUT

A terminal window with a dark background. The title bar shows three colored circles (red, yellow, green) followed by the text "Five.java". The prompt is "> java Five" in green. The output is "invalid type casting" in white.

```
Five.java
> java Five
invalid type casting
```

IllegalArgumentException

Six.java

CODE

```
package exception;

public class Six {

    static void setAge(int age) {

        if (age<0) {

            throw new IllegalArgumentException("Age cannot be negative");

        }

        System.out.println("Age:"+age);
    }

    public static void main(String[] args) {

        try {

            setAge(20);
            setAge(-5);

        }

        catch (IllegalArgumentException e) {

            System.out.println("Error: " + e.getMessage());

        }

        System.out.println("Program continues");

    }

}
```

OUTPUT

● ● ● Six.java

> java Six

Age:20

Error: Age cannot be negative

Program continues

IllegalStateException

Seven.java

CODE

```
package exception;

class Printer{

    boolean on = false;

    void print() {

        if (!on) {

            throw new IllegalStateException("Printer is turned off");

        }

        System.out.println("Printing...");

    }

}

public class Seven {

    public static void main(String[] args) {

        Printer p = new Printer();

        try {

            p.print();

        }

        catch (IllegalStateException e) {

            System.out.println(e.getMessage());

        }

    }

}
```

OUTPUT

Seven.java

> java Seven

Printer is turned off

NegativeArraySizeException

Eight.java

CODE

```
package exception;

public class Eight {

    public static void main(String[] args) {

        try {

            //size cannot be negative it has to be either 0 or 1,2,-----

            int size = -5;

            int[] numbers = new int[size];

            System.out.println("Array created");

        }

        catch (NegativeArraySizeException e) {

            System.out.println("Array size cannot be negative");

        }

    }

}
```

OUTPUT

Eight.java

> **java Eight**

Array size cannot be negative

UnsupportedOperationException

Nine.java

CODE

```
package exception;

import java.util.Arrays;
import java.util.List;

public class Nine {

    public static void main(String[] args) {

        List<Integer> list = Arrays.asList(10, 20, 30);

        try {

            list.add(40);

        }

        catch (UnsupportedOperationException e) {

            System.out.println("Add operation is not supported");

        }

    }

}
```

OUTPUT

● ● ● Nine.java

> **java Nine**

Add operation is not supported

ArrayStoreException

Ten.java

CODE

```
package exception;

public class Ten {

    public static void main(String[] args) {

        Object[] arr = new String[3];

        try {

            arr[0] = "Java";

            arr[1] = "Python";

            arr[2] = 10;//it is an integer

        }

        catch (ArrayStoreException e) {

            System.out.println(e.getMessage());

        }

    }

}
```

OUTPUT

A terminal window with a dark background and a title bar containing three colored circles (red, yellow, green) and the text "Ten.java". The terminal shows a green prompt character followed by the command "java Ten". The output of the command is "java.lang.Integer".

```
Ten.java

> java Ten
java.lang.Integer
```